

PRINCE CHAUHAN

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Phone: +91-9265212936 | **Patents:** IN 202441065103, IN 202441065106

PROFESSIONAL SUMMARY

Thermal Engineer with 3+ years of experience designing Battery Thermal Management Systems (BTMS) and Charger Thermal Management Systems (CTMS) for 20–110 kWh battery swapping and fast-charging platforms operating under extreme ambient conditions (up to 55°C). Experienced in first-principles thermal system design, including heat load estimation, thermal analysis, coolant flow sizing, radiator/chiller selection, and system-level integration. Proven expertise in thermal validation, failure analysis, and developing reliable, lightweight, and high-performance thermal architectures for high-power energy systems. Interested in applying strong heat transfer and thermal engineering fundamentals to high-reliability aerospace systems, including spacecraft and satellite thermal control.

TECHNICAL SKILLS

Thermal Design & Analysis

Battery Thermal Management Systems (BTMS), Charger Thermal Management Systems (CTMS), Liquid-Cooled Radiator Systems, High-Power Electronics Cooling, Heat Transfer, Heat Load Estimation, Thermal System Design, Thermal System Integration, Thermal Runaway Prevention, Multi-Scale Thermal Modeling (Cell/Pack/Station)

Computational & Simulation Tools

CFD: ANSYS Fluent, Flow Simulation (coolant dynamics, heat transfer analysis) | CAD: SolidWorks, CREO (thermal system design)

Testing & Validation

Environmental Chamber Testing (–10°C to 60°C), Thermal Sensor Calibration (RTD, Thermocouples), IR Thermography, Data Acquisition Systems, Root Cause Analysis (RCA), Design of Experiments (DOE), Factory Validation Protocols, Supplier Quality Management

Embedded Systems & Control

ECU Simulation, Battery Management System (BMS) Architecture, State Machine Design, Alert Logic Development, Real-time Thermal Monitoring, CAN Protocol

WORK EXPERIENCE

Senior Thermal Systems Engineer | Sun Mobility, Bangalore (Heavy Electric Vehicles Dept) | 2023–Present

Supplier Collaboration & Factory Validation

Partnered with Chinese battery manufacturers to optimize thermal efficiency for 20-110 kWh battery swapping systems, reducing charging thermal resistance by 18% through iterative cooling design improvements.

- Led thermal validation at OEM factories including test setup design, data acquisition protocol development, thermal performance reporting, and training factory engineers on validation procedures for multi-pack charging stations.

Thermal System Design & Testing

- Designed and integrated Battery Thermal Management Systems (BTMS) and Charger Thermal Management Systems (CTMS) for 20–110 kWh battery swapping and fast-charging platforms operating under ambient temperatures up to 55°C. Defined the complete thermal architecture from first principles, including heat load estimation, coolant flow sizing, radiator/chiller selection, thermal limits, and system-level integration to ensure reliable operation under extreme duty cycles.
- Architected and validated coolant thermal management systems for the **Prawaas 4.0 Auto Expo** demonstration, maintaining **650 kg battery packs** within **25–35°C** during **110 kWh supercharging** under **50°C ambient conditions**, enabling **100+ successful battery swaps**.
- Calculated battery and charger heat loads using first-principles energy balance and loss modeling, sizing radiators, chillers, and coolant flow rates based on required heat rejection and ΔT constraints.
- Optimized radiator sizing and coolant loop performance through flow distribution analysis and thermal efficiency improvements under peak charging loads (1.6C–3C).
- Authored System of Requirements (SOR) documents defining BTMS/CTMS state machines, flow diagrams, and alert logic adopted as governing control logic by embedded systems team
- Developed ECU–station controller simulator and battery over-charger cooling safety logic for validation and stress-testing of control strategies under extreme ambient conditions

Innovation & Patents

- Inventor on 2 patent applications: thermal management of chargers (IN 202441065103) and electrical energy storage devices (IN 202441065106), focusing on intelligent adaptive cooling strategies
- Recognized with InvenZone Award (2025) for patent contributions to EV thermal management innovations

Simulation Engineer | GTU Motorsports Team, Formula Bharat | 2018–2022

- Led chassis design optimization using CAD and FEA simulation, reducing vehicle weight by 15%, contributing to national victory against 80+ teams including IITs and NITs in Formula Bharat Competition (2019)
- Performed structural analysis (stress, strain, deformation, and factor of safety) on race car chassis components using FEA, supporting lightweight design under severe performance constraints.

ACADEMIC PROJECTS

Solar Tree System Design | Project Engineer | 2022–2023

- Designed 3 kW solar tree system using SolidWorks for off-grid/on-grid deployment, reducing costs 50% (\$4,000 → \$2,000) and spatial footprint 80% vs conventional solar panel systems
- Ensured 25-year design lifespan with structural resilience to 90 km/hr wind speeds through thermal-structural coupled analysis and material selection optimization.

EDUCATION

B.E. in Automobile Engineering

L.J. Institute of Engineering & Technology (LJIET), Gujarat, India | Graduated 2022

Relevant Coursework: Heat Transfer, Thermodynamics, Fluid Mechanics, Computational Methods, Vehicle Thermal Management

CERTIFICATIONS & PROFESSIONAL DEVELOPMENT

- Aerospace Engineering: Astronautics and Human Spaceflight — Massachusetts Institute of Technology (MITx)
- Machine Learning Specialization — Andrew Ng, Coursera (Supervised Learning, Neural Networks, Predictive Modeling)
- IBM Data Science Professional Certificate — Coursera (Data Analysis, Python for Data Science, ML Workflows)
- Electric Vehicle Technology — Delft University of Technology (DelftX)
- Solar Energy Systems — Delft University of Technology (DelftX)

SELECTED ACHIEVEMENTS & AWARDS

InvenZone Award for Patent Contributions | Sun Mobility, Bangalore | 2025

Recognized for multiple patent-pending innovations in EV thermal management and charging systems for multi-battery-pack charging stations

National Winner, Formula Bharat Competition | GTU Motorsports Team | 2019

Led team to 1st place in endurance run, outperforming 80+ teams including top institutions (IITs, NITs) through innovative chassis design and constraint-based optimization